

A. Introduction

1. **Title:** Cyber Security — Physical Security
2. **Number:** RRC-CIP-014-1
3. **Purpose:** To identify and protect Transmission Facilities, and their associated primary control centers, that if rendered inoperable or damaged as a result of a physical attack could result in instability, uncontrolled separations, or cascading outages within the Bulk Electric System (BES).

4. Applicability:

4.1. Functional Entities:

4.1.1. Transmission Owner that owns a Transmission Facility that meets any of the following criteria:

4.1.1.1. Transmission Facilities operated at 300 kV or higher. For this purpose of this criterion, the collector bus for a generation plant is not considered a Transmission Facility but is part of the generation interconnection Facility.

4.1.1.2. Transmission Facilities that are operating between 100 kV and 299 kV at a single station or substation, where the station or substation is connected at 100 kV or higher voltages to three or more other Transmission Facilities and has an "aggregate weighted value" exceeding 2700 according to the table below. The "aggregate weighted value" for a single station or substation is determined by summing the "weight value per line" shown in the table below for each incoming and each outgoing BES Transmission Line that is connected to another Transmission Facility. For the purpose of this criterion, the collector bus for a generation plant is not considered a Transmission Facility but is part of the generation

Voltage Value of a Line	Weight Value per Line
less than 100 kV (not applicable)	(not applicable)
100 kV to 199 kV	700
200 kV to 299 kV	1300

4.1.1.3. Transmission Facilities at a single station or substation location that are identified by the Planning Authority as critical to the derivation of Interconnection Reliability Operating Limits (IROLs) and their associated contingencies.

4.1.1.4. Transmission Facilities providing the generation interconnection required to connect generator output to the Transmission Systems that, if destroyed, degraded, misused, or otherwise rendered unavailable, would result in the loss of the generation Facilities identified by any Generator Owner as a result of its application of Attachment 1, criterion 2.1 or 2.3.

4.1.2. Transmission Operator

Exemptions: Facilities in a "protected area," as defined in 10 C.F.R. § 73.2, within the scope of a security plan approved or accepted by the Nuclear Regulatory Commission are not subject to this

Standard; or, Facilities within the scope of a security plan approved or accepted by the Canadian Nuclear Safety Commission are not subject to this Standard.

5. **Effective Dates:** Shall become effective on the first day of the first calendar quarter that is thirty-six (36) months following adoption by the Approving Authority.

B. Requirements and Measures

- R1.** Each Transmission Owner shall perform an initial risk assessment and subsequent risk assessments of Railbelt Transmission Facilities (existing and planned to be in service within 24 months) that meet the criteria specified in Applicability Section 4.1.1. The risk assessment shall be conducted following the steps outlined in RRC-CIP-014-1 Attachment 1. The initial and subsequent risk assessments shall consist of a transmission analysis or transmission analyses designed to identify the Transmission Facility(s) that if rendered inoperable or damaged, could result in instability, uncontrolled separations, or cascading outages within an Interconnection. *[Violation Risk Factor: High]*
- 1.1.** Subsequent risk assessments shall be performed:
- At least once every 42 calendar months for a Transmission Owner that has identified in its previous risk assessment (as verified according to Requirement R2) one or more Transmission facility that if rendered inoperable or damaged could result in instability, uncontrolled separations, or cascading outages within an Interconnection; or
 - At least once every 60 calendar months for a Transmission Owner that has not identified in its previous risk assessment (as verified according to Requirement R2) any Transmission facility that if rendered inoperable or damaged could result in instability, uncontrolled separations, or cascading outages within an Interconnection.
 - If evaluation of past events is used as a methodology by a Transmission Owner, these assessments shall be submitted within 12 calendar months of the event occurrence.
- 1.2.** The Transmission Owner shall identify the primary control center that operationally controls each Transmission facility identified in the Requirement R1 risk assessment.
- M1.** Examples of acceptable evidence may include, but are not limited to, dated written or electronic documentation of the risk assessment results identifying the list of Transmission facilities that, if rendered inoperable or damaged, could result in instability, uncontrolled separations, or cascading outages within an Interconnection.
- R2.** The Planning Authority will perform a review of the risk assessment that was completed as part of the requirement of R1
- 2.1.** Any issues or suggested changes to the assessment shall be provided in writing, with evidence to support changes to the risk assessment, to the Transmission Owner for review and disposition.
- 2.2.** Transmission Owner, after review and disposition, shall issue a new risk assessment as per R1 if applicable.
- M2.** Examples of acceptable evidence may include, but are not limited to, dated written or electronic documentation from the Planning Authority that they completed a verification of the risk assessment requirement of R1

- R3.** For a primary control center(s) identified by the Transmission Owner according to Requirement R1, Part 1.2 that a) operationally controls an identified Transmission Facility verified according to Requirement R2, and b) is not under the operational control of the Transmission Owner: the Transmission Owner shall, within seven calendar days following completion of Requirement R2, notify the Transmission Operator that has operational control of the primary control center of such identification and the date of completion of Requirement R2. [VRF: Lower; Time-Horizon: Long-term Planning]
- 3.1.** If a Transmission Facility previously identified under Requirement R1 and verified according to Requirement R2 is removed from the identification during a subsequent risk assessment performed according to Requirement R1 or a verification according to Requirement R2, then the Transmission Owner shall, within seven calendar days following the verification or the subsequent risk assessment, notify the Transmission Operator that has operational control of the primary control center of the removal.
- M3.** Examples of acceptable evidence may include, but are not limited to, dated written or electronic notifications or communications that the Transmission Owner notified each Transmission Operator, as applicable, according to Requirement R3.
- R4.** Each Transmission Owner that identified a Transmission Facility, or a primary control center in Requirement R1 and verified according to Requirement R2, and each Transmission Operator notified by a Transmission Owner according to Requirement R3, shall conduct an evaluation of the potential threats and vulnerabilities of a physical attack to each of their respective Transmission facilities and primary control center(s) identified in Requirement R1 and verified according to Requirement R2. The evaluation shall consider the following: [*Violation Risk Factor: Medium*]
- 4.1.** Unique characteristics of the identified and verified Transmission Facility(s), and primary control center(s);
- 4.2.** Prior history of attack on similar facilities taking into account the frequency, geographic proximity, and severity of past physical security related events; and
- 4.3.** Intelligence or threat warnings received from sources such as law enforcement, the Electric Reliability Organization (ERO), the Electricity Information Sharing and Analysis Center (E-ISAC), U.S. federal and/or Canadian governmental agencies, or their successors.
- M4.** Examples of evidence may include, but are not limited to, dated written or electronic documentation that the Transmission Owner or Transmission Operator conducted an evaluation of the potential threats and vulnerabilities of a physical attack to their respective Transmission Facility (s) and primary control center(s) as specified in Requirement R4.
- R5.** Each Transmission Owner that identified a Transmission Facility or primary control center in Requirement R1 and verified according to Requirement R2, and each Transmission Operator notified by a Transmission Owner according to Requirement R3, shall develop and implement a documented physical security plan(s) that covers their respective Transmission Facility(s), and primary control center(s). The physical security plan(s) shall be developed within 120 calendar days following the completion of Requirement R2 and executed according to the timeline specified in the physical security plan(s). The physical security plan(s) shall include the following attributes: [*Violation Risk Factor: High*]

- 5.1.** Resiliency or security measures designed collectively to deter, detect, delay, assess, communicate, and respond to potential physical threats and vulnerabilities identified during the evaluation conducted in Requirement R4.
 - 5.2.** Law enforcement contact and coordination information.
 - 5.3.** A timeline for executing the physical security enhancements and modifications specified in the physical security plan.
 - 5.4.** Provisions to evaluate evolving physical threats, and their corresponding security measures, to the Transmission Facility(s), or primary control center(s).
- M5.** Examples of evidence may include, but are not limited to, dated written or electronic documentation of its physical security plan(s) that covers their respective identified and verified Transmission Facility(s), and primary control center(s) as specified in Requirement R5, and additional evidence demonstrating execution of the physical security plan according to the timeline specified in the physical security plan.
- R6.** Each Transmission Owner and Transmission Operator shall have their physical security plan(s) reviewed by the RRC for conformance with the requirements of CIP-014. The plan(s) must be submitted to the RRC, which will review them and provide feedback on any recommended changes or concerns. *[Violation Risk Factor: Medium]*
 - 6.1.** If the RRC recommends changes to the evaluation performed under Requirement R4 or security plan(s) developed under Requirement R5, the Transmission Owner or Transmission Operator shall, within 60 calendar days of the completion of the review, for each recommendation:
 - Modify its evaluation or security plan(s) consistent with the recommendation; or
 - Document the reason(s) for not modifying the evaluation or security plan(s) consistent with the recommendation.
- M6.** Examples of evidence may include, but are not limited to, written or electronic documentation that the Transmission Owner or Transmission Operator documented the reasons for not modifying the evaluation or security plan(s) in accordance with a recommendation under Part 6.3. Additionally, examples of evidence may include, but are not limited to, written or electronic documentation of procedures to protect information under Part 6.4.

C. Compliance

1. Compliance Monitoring Process

1.1 Compliance Enforcement Authority:

The RRC is the entity with the authority to monitor and enforce compliance with mandatory and enforceable Railbelt Reliability Council Standards.

1.2 Evidence Retention:

The following evidence retention periods identify the period of time an entity is required to retain specific evidence to demonstrate compliance. For instances where the evidence retention period specified below is shorter than the time since the last audit, the RRC may ask an entity to provide other evidence to show that it was compliant for the full time period since the last audit.

The Responsible Entity shall keep data or evidence to show compliance as identified below unless directed by the RRC to retain specific evidence for a longer period of time as part of an investigation:

- Each Responsible Entity shall retain evidence of each requirement in this standard for three calendar years.
- If a Responsible Entity is found non-compliant, it shall keep information related to the non-compliance until mitigation is complete and approved or the time specified above, whichever is longer.
- The RRC shall keep the last audit records and all requested and submitted subsequent audit records.

1.3 Compliance Monitoring and Enforcement Program:

Compliance Monitoring and Enforcement Program refers to the identification of the processes that will be used to evaluate data or information for the purpose of assessing performance or outcomes with the associated Railbelt Reliability Council Standards.

Table of Compliance Elements

R#	Violation Risk Factor (VRF)	Violation Severity Levels (RRC-CIP-014-1)			
		Lower VSL	Moderate VSL	High VSL	Severe VSL
R1	High	The Transmission Owner performed an initial risk assessment but did so after the enforcement date for performing the initial risk assessment but less than or equal to one calendar months after that date; OR The Transmission Owner that has identified in its previous risk assessment one or more Transmission Facilities that if rendered inoperable or damaged could result in instability, uncontrolled separation, or Cascading within an Interconnection performed a subsequent risk assessment but did so after 42 calendar months but less than or equal to 43 calendar months; OR	The Transmission Owner performed an initial risk assessment but did so more than one calendar months after the enforcement date for performing the initial risk assessment but less than or equal to two calendar months after that date; OR The Transmission Owner that has identified in its previous risk assessment one or more Transmission Facilities that if rendered inoperable or damaged could result in instability, uncontrolled separation, or Cascading within an Interconnection performed a subsequent risk assessment but did so after 43 calendar months but less than or equal to 44 calendar months;	The Transmission Owner performed an initial risk assessment but did so more than two calendar months after the enforcement date for performing the initial risk assessment but less than or equal to three calendar months after that date; OR The Transmission Owner that has identified in its previous risk assessment one or more Transmission Facilities that if rendered inoperable or damaged could result in instability, uncontrolled separation, or Cascading within an Interconnection performed a subsequent risk assessment but did so after 44 calendar months but less than or equal to 45 calendar months;	The Transmission Owner performed an initial risk assessment but did so more than three calendar months after the enforcement date for performing the initial risk assessment; OR The Transmission Owner failed to perform an initial risk assessment; OR The Transmission Owner that has identified in its previous risk assessment one or more Transmission Facilities that if rendered inoperable or damaged could result in instability, uncontrolled separation, or Cascading within an Interconnection performed a subsequent risk assessment but did so after

R#	Violation Risk Factor (VRF)	Violation Severity Levels (RRC-CIP-014-1)			
		Lower VSL	Moderate VSL	High VSL	Severe VSL
		The Transmission Owner that has not identified in its previous risk assessment any Transmission Facilities that if rendered inoperable or damaged could result in instability, uncontrolled separation, or Cascading within an Interconnection performed a subsequent risk assessment but did so after 60 calendar months but less than or equal to 61 calendar months.	OR The Transmission Owner that has not identified in its previous risk assessment any Transmission Facilities that if rendered inoperable or damaged could result in instability, uncontrolled separation, or Cascading within an Interconnection performed a subsequent risk assessment but did so after 61 calendar months but less than or equal to 62 calendar months.	OR The Transmission Owner that has not identified in its previous risk assessment any Transmission Facilities that if rendered inoperable or damaged could result in instability, uncontrolled separation, or Cascading within an Interconnection performed a subsequent risk assessment but did so after 62 calendar months but less than or equal to 63 calendar months; OR The Transmission Owner performed a risk assessment but failed to include R 1.2.	more than 45 calendar months; OR The Transmission Owner that has identified in its previous risk assessment one or more Transmission Facilities that if rendered inoperable or damaged could result in instability, uncontrolled separation, or Cascading within an Interconnection failed to perform a risk assessment; OR The Transmission Owner that has not identified in its previous risk assessment any Transmission Facilities that if rendered inoperable or damaged could result in instability, uncontrolled separation, or Cascading within an Interconnection performed a subsequent risk assessment but did so

R#	Violation Risk Factor (VRF)	Violation Severity Levels (RRC-CIP-014-1)			
		Lower VSL	Moderate VSL	High VSL	Severe VSL
					after more than 63 calendar months; OR The Transmission Owner that has not identified in its previous risk assessment any Transmission station and Transmission substations that if rendered inoperable or damaged could result in instability, uncontrolled separation, or Cascading within an Interconnection failed to perform a subsequent risk assessment.
R2	Medium	The Transmission Owner issued a revised R1 risk assessment, if applicable, more than thirty calendar days but no more than forty calendar days after receiving the Planning Authority's comments.	The Transmission Owner issued a revised R1 risk assessment, if applicable, more than forty calendar days but no more than fifty calendar days after receiving the Planning Authority's comments.	The Transmission Owner issued a revised R1 risk assessment, if applicable, more than fifty calendar days but no more than sixty calendar days after receiving the Planning Authority's comments OR the Planning Authority failed to include sufficient supporting evidence in its written feedback, as required by R 2.1.	The Transmission Owner failed to review and disposition the Planning Authority's feedback OR the Transmission Owner did not issue a revised R1 risk assessment when the feedback required changes.

R#	Violation Risk Factor (VRF)	Violation Severity Levels (RRC-CIP-014-1)			
		Lower VSL	Moderate VSL	High VSL	Severe VSL
R3	Lower	The Transmission Owner notified the Transmission Operator that operates the primary control center as specified in Requirement R3 but did so more than seven calendar days and less than or equal to 22 calendar days following the completion of Requirement R2; OR The Transmission Owner notified the Transmission Operator that operates the primary control center of the removal from the identification in Requirement R1 but did so more than seven calendar days and less than or equal to 22 calendar days following the verification or the subsequent risk assessment.	The Transmission Owner notified the Transmission Operator that operates the primary control center as specified in Requirement R3 but did so more than 22 calendar days and less than or equal to 37 calendar days following the completion of Requirement R2; OR The Transmission Owner notified the Transmission Operator that operates the primary control center of the removal from the identification in Requirement R1 but did so more than 22 calendar days and less than or equal to 37 calendar days following the verification or the subsequent risk assessment.	The Transmission Owner notified the Transmission Operator that operates the primary control center as specified in Requirement R3 but did so more than 37 calendar days and less than or equal to 52 calendar days following the completion of Requirement R2; OR The Transmission Owner notified the Transmission Operator that operates the primary control center of the removal from the identification in Requirement R1 but did so more than 37 calendar days and less than or equal to 52 calendar days following the verification or the subsequent risk assessment.	The Transmission Owner notified the Transmission Operator that operates the primary control center as specified in Requirement R3 but did so more than 52 calendar days following the completion of Requirement R2; OR The Transmission Owner failed to notify the Transmission Operator that it operates a control center identified in Requirement R1; OR The Transmission Owner notified the Transmission Operator that operates the primary control center of the removal from the identification in Requirement R1 but did so more than 52 calendar days following the verification or

R#	Violation Risk Factor (VRF)	Violation Severity Levels (RRC-CIP-014-1)			
		Lower VSL	Moderate VSL	High VSL	Severe VSL
					the subsequent risk assessment. OR The Transmission Owner failed to notify the Transmission Operator that operates the primary control center of the removal from the identification in Requirement R1.
R4	Medium	N/A	The Responsible Entity conducted an evaluation of the potential physical threats and vulnerabilities to each of its Transmission Facility(s), and primary control center(s) identified in Requirement R1 but failed to consider one of Parts in R 4.1 through R 4.3 in the evaluation.	The Responsible Entity conducted an evaluation of the potential physical threats and vulnerabilities to each of its Transmission Facility(s), and primary control center(s) identified in Requirement R1 but failed to consider two of Parts in R 4.1 through R 4.3 in the evaluation.	The Responsible Entity failed to conduct an evaluation of the potential physical threats and vulnerabilities to each of its Transmission Facility(s), and primary control center(s) identified in Requirement R1; OR The Responsible Entity conducted an evaluation of the potential physical threats and vulnerabilities to each of its Transmission

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R#	Violation Risk Factor (VRF)	Violation Severity Levels (RRC-CIP-014-1)			
		Lower VSL	Moderate VSL	High VSL	Severe VSL
					Facility(s), and primary control center(s) identified in Requirement R1 but failed to consider any Part in R 4.1 through R 4.3.
R5	High	The Responsible Entity developed and implemented a documented physical security plan(s) that covers each of its Transmission Facility(s), and primary control center(s) identified in Requirement R1 but did so more than 120 calendar days but less than or equal to 125 calendar days after completing Requirement R2; OR The Responsible Entity developed and	The Responsible Entity developed and implemented a documented physical security plan(s) that covers each of its Transmission Facility(s), and primary control center(s) identified in Requirement R1 but did so more than 125 calendar days but less than or equal to 130 calendar days after completing Requirement R2; OR The Responsible Entity developed and	The Responsible Entity developed and implemented a documented physical security plan(s) that covers each of its Transmission Facility(s), and primary control center(s) identified in Requirement R1 but did so more than 130 calendar days but less than or equal to 135 calendar days after completing Requirement R2; OR The Responsible Entity developed and	The Responsible Entity developed and implemented a documented physical security plan(s) that covers each of its Transmission Facility(s), and primary control center(s) identified in Requirement R1 but did so more than 135 calendar days after completing the verification in Requirement R2; OR The Responsible Entity failed to develop and implement a documented

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R#	Violation Risk Factor (VRF)	Violation Severity Levels (RRC-CIP-014-1)			
		Lower VSL	Moderate VSL	High VSL	Severe VSL
		implemented a documented physical security plan(s) that covers its Transmission Facility(s), and primary control center(s) identified in Requirement R1 and verified according to Requirement R2 but failed to include one of R 5.1 through R 5.4 in the plan.	implemented a documented physical security plan(s) that covers its Transmission Facility(s), and primary control center(s) identified in Requirement R1 and verified according to Requirement R2 but failed to include two of R 5.1 through R 5.4 in the plan.	implemented a documented physical security plan(s) that covers its Transmission Facility(s), and primary control center(s) identified in Requirement R1 and verified according to Requirement R2 but failed to include three of R 5.1 through R 5.4 in the plan.	physical security plan(s) that covers its Transmission Facility(s), and primary control center(s) identified in Requirement R1 and verified according to Requirement R2. OR The Responsible Entity developed and implemented a documented physical security plan(s) that covers its Transmission Facility(s), and primary control center(s) identified in Requirement R1 and verified according to Requirement 2 but failed to include R 5.1 through R 5.4 in the plan.

R#	Violation Risk Factor (VRF)	Violation Severity Levels (RRC-CIP-014-1)			
		Lower VSL	Moderate VSL	High VSL	Severe VSL
R6	Medium	The Responsible Entity had RRC review the evaluation performed under Requirement R4 and the security plan(s) developed under Requirement R5 and modified or documented the reason for not modifying the security plan(s) as specified in Part 6.1 but did so more than 60 calendar days and less than or equal to 70 calendar days following completion of the RRC review.	The Responsible Entity had RRC review the evaluation performed under Requirement R4 and the security plan(s) developed under Requirement R5 and modified or documented the reason for not modifying the security plan(s) as specified in Part 6.1 but did so more than 70 calendar days and less than or equal to 80 calendar days following completion of the RRC review.	The Responsible Entity had RRC review the evaluation performed under Requirement R4 and the security plan(s) developed under Requirement R5 and modified or documented the reason for not modifying the security plan(s) as specified in Part 6.1 but did so more than 80 calendar days and less than or equal to 90 calendar days following completion of the RRC review.	The Responsible Entity failed to have an RRC review the evaluation performed under Requirement R4 and the security plan(s) developed under Requirement R5; OR The Responsible Entity RRC reviewed the evaluation performed under Requirement R4 and the security plan(s) developed under Requirement R5, but failed to modify its evaluation or security plans or document the reasons for not modifying the evaluation of security plans

D. Regional Variances

None.

E. Interpretations

None.

F. Associated Documents

None.

Version History

Version	Date	Action	Change Tracking

Attachment 1

Guideline to Perform Transmission Risk Assessment

- 1. Step One:** The Transmission Owner identifies stations to be analyzed based on applicability criteria 4.1.1.
 - 1.1.** Note 1: Stations are both existing stations as well as those planned to be in service within 24 months.
 - 1.2.** Note 2: In performing this analysis, the general approach is to take out one station at a time, not a combination of stations. A Transmission Owner may determine it is appropriate to take out more than one station at a time, as a result of two or more stations being in close proximity to one another. An example of the type of factors to consider, when considering close proximity, is where proximity is defined as having two (or more) substations situated such that there is either (i) an easy line-of-sight between all of the substation yards from a single site, (ii) an easy access from a common public roadway that exists between all of the substation yards, or (iii) the substation yards are in close enough proximity that a single event can impact both substations (e.g., the debris field from a reasonable incendiary device set off at one yard will impact the other yard). If such conditions exist, consider grouping these substations together before proceeding and treat them as a single substation when performing the next step.
- 2. Step Two:** The Transmission Owner identifies cases/system conditions to be analyzed. Cases/system conditions should represent stressful system conditions on the transmission system based on the engineering knowledge and judgment of the planner performing the actual studies for that Transmission Owner's system. Possible items for consideration are:
 - 2.1.** summer peak vs. winter peak load levels
 - 2.2.** shoulder peak load levels with system transfers
 - 2.3.** alternative generation dispatch assumptions
 - 2.4.** alternative load models (i.e., different penetration of inductive load)
- 3. Step Three:** The Transmission Owner defines the nature of the initiating event and how it will be modeled in the transmission assessment. Possible items for consideration are:
 - 3.1.** an event that evolves over several minutes allowing time for system operator intervention
 - 3.2.** an effectively instantaneous event involving an explosive or other incendiary device that would result in a fault in the station under attack, which would result in the operation of protective relays to remove the station under attack from the system
- 4. Step Four:** The Transmission Owner is responsible for the development of criteria/proxies for instability, uncontrolled separation or Cascading, based on the engineering knowledge and judgment of its system. In developing the criteria for the RRC-CIP-014-1 R1 assessment, the following can be considered:
 - 4.1.** applicable regional guidelines, if any
 - 4.2.** post-contingency overload percentage, above applicable ratings, for the Transmission Owner's utility/region

- 4.3. post-contingency voltages or voltage deviation
- 4.4. amount of load loss
- 4.5. amount of generation loss, including those generating units lost due to instability
- 4.6. if stability simulations are performed, the following can be considered:
 - the transient voltage response, which includes both the magnitude and duration of voltage excursion
 - negatively damped oscillations/poorly damped or undamped oscillations
 - tripping of lines because of apparent impedance
 - frequency excursions

5. Step Five: The Transmission Owner performs appropriate steady-state power flow and/or stability analysis. As the Transmission Owner develops its specific process for performing steady-state and/or stability analysis, the Transmission Owner may consider incorporating the following concepts:

5.1. Steady-state power flow analysis

- In performing the steady-state power flow analysis, consider an outage of the entire station as identified in step one and examine the immediate response of the power system to the loss of that substation. That is, all buses in a physically contiguous station are isolated from the transmission system remotely by opening corresponding breakers according to an actual breaker diagram. Upon utility discretion, events at stations, already identified under other planning studies that do not follow this assumption but cause greater impact, can be substituted for assuming all lines into a substation are removed (e.g., a situation where loss of part of the station as opposed to the entire station creates a more severe system impact).
- In performing the steady-state analysis, the Transmission Owner needs to define what sort of “operator action/planner discretion” will be allowed at each step of the analysis, which would result in the elimination/screening out of a station as a RRC-CIP-014-1 R1 “critical facility” station. Examples of “operator action/planner discretion” are:
 - When all loading is below Transmission Owner defined acceptable threshold for assuming operator action to mitigate overloads, the station may be screened out. Typical limitations on mitigation measures imposed under TPL standards (such as loss of local load) do not apply.
 - Event cascades but is limited in scope to a defined area of acceptable size (i.e., is determined to not have a critical impact on the operation of the Interconnection).
 - Event cascades but if project(s) (e.g., ancillary equipment upgrade) can and will be quickly initiated to eliminate/mitigate scope of cascading to an area of acceptable size.
- For simulations where solution convergence is obtained, lines found to overload beyond their acceptable threshold limits (e.g., including protection limits if known) will be taken out of service and a new solution will be attempted. This process will be repeated until either no lines overload past their acceptable threshold limits or the process has been repeated a defined number of times, as determined by the Transmission Owner, and continues to result in lines overloading past their acceptable threshold limits (i.e., Cascading).
- Bus voltages assessed as defined in Step Four above.

- Amount of load loss assessed as defined in Step Four above.
- Amount of generation loss assessed as defined in Step Four above.
- Analysis should include monitoring of facilities in systems beyond the system in which the station being analyzed is located. For cascading beyond a Transmission Owner area, neighboring Transmission Owners should work together as necessary to determine the extent of the cascading event(s).
- For simulations where solution convergence is not obtained, the Transmission Owner may want to consider further analysis to determine the cause of the non-convergence.

5.2. Stability analysis (if performed)

- If performing stability analysis, the analysis needs to consider the following:
 - The nature of the fault in the station(s) being analyzed (e.g., single phase fault, 3-phase fault, etc.).
 - The placement of the fault in the station(s) being analyzed (e.g., fault located on highest voltage bus of station being tested, fault located on all busses in station being tested, etc.).
 - How the fault will be cleared (e.g., normal clearing by breakers in station being analyzed, remote clearing with normal clearing time, remote clearing with time delays due to loss of relay communication channels, etc.).
- Generator rotor angles assessed for potential transient instability issues as defined in Step Four above.
- Oscillation response assessed for negatively damped oscillations/poorly damped or undamped oscillations as defined in Step Four above.
- Transient voltage response (if applicable) assessed per transient voltage criteria as defined in Step Four above.
- Assessment of other stability criteria as defined in Step Four above.
- In assessing the stability analysis results, results should be viewed as stated in both the March 17, 2014, Order (paragraph 6), the July 17, 2014, NOPR (paragraph 25) and the November 20, 2014, Order (paragraph 33) to assess if the loss of the specific Transmission Facility qualifies as a “critical facility” where: “A critical facility is one that, if rendered inoperable or damaged, could have a critical impact on the operation of the interconnection through instability, uncontrolled separation or cascading failures on the Bulk-Power System.” The instability of a single generator or multiple generators due to the loss of a Transmission Facility does not necessarily mean that that Transmission Facility is a “critical facility.” Rather, the threshold for “critical facility” is tied to how the loss of the Transmission Facility impacts the broader Interconnection.

Glossary

Glossary

Balancing Authority	The responsible entity that integrates resource plans ahead of time, maintains Demand and resource balance within a Balancing Authority Area, and supports Interconnection frequency in real time.
BES Cyber Asset (BCA)	A Cyber Asset that if rendered unavailable, degraded, or misused would, within 15 minutes of its required operation, misoperation, or non operation, adversely impact one or more Facilities, systems, or equipment, which, if destroyed, degraded, or otherwise rendered unavailable when needed, would affect the reliable operation of the Bulk Electric System. Redundancy of affected Facilities, systems, and equipment shall not be considered when determining adverse impact. Each BES Cyber Asset is included in one or more BES Cyber Systems.
BES Cyber System (BCS)	One or more BES Cyber Assets logically grouped by a responsible entity to perform one or more reliability tasks for a functional entity.
BES Cyber System Information	Information about the BES Cyber System that could be used to gain unauthorized access or pose a security threat to the BES Cyber System. BES Cyber System Information does not include individual pieces of information that by themselves do not pose a threat or could not be used to allow unauthorized access to BES Cyber Systems, such as, but not limited to, device names, individual IP addresses without context, ESP names, or policy statements. Examples of BES Cyber System Information may include, but are not limited to, security procedures or security information about BES Cyber Systems, Physical Access Control Systems, and Electronic Access Control or Monitoring Systems that is not publicly available and could be used to allow unauthorized access or unauthorized distribution; collections of network addresses; and network topology of the BES Cyber System.
Blackstart Resource	A generating unit(s) and its associated set of equipment which has the ability to be started without support from the System or is designed to remain energized without connection to the remainder of the System, with the ability to energize a bus, meeting the Transmission Operator's restoration plan needs for Real and Reactive Power capability, frequency and voltage control, and that has been included in the Transmission Operator's restoration plan.
Bulk Electric System (BES)	Unless modified by the lists shown below, all Transmission Elements operated at 100 kV or higher and Real Power and Reactive Power resources connected at 100 kV or higher. This does not include facilities used in the local distribution of electric energy. Inclusions: • I1 - Transformers with the primary terminal and at least one secondary terminal operated at 100 kV or higher unless excluded by application of Exclusion E1 or E3.

	• I2 – Generating resource(s) including the generator terminals through the high-side of the step-up transformer(s) connected at a voltage of 100 kV or above with: a) Gross individual nameplate rating greater than 20 MVA. Or, b) Gross plant/facility aggregate nameplate rating greater than 75 MVA. • I3 - Blackstart Resources identified in the Transmission Operator's restoration plan. • I4 - Dispersed power producing resources that aggregate to a total capacity greater than 75 MVA (gross nameplate rating), and that are connected through a system designed primarily for delivering such capacity to a common point of connection at a voltage of 100 kV or above. Thus, the facilities designated as BES are: a) The individual resources, and The system designed primarily for delivering capacity from the point where those resources aggregate to greater than 75 MVA to a common point of connection at a voltage of 100 kV or above. • I5 –Static or dynamic devices (excluding generators) dedicated to supplying or absorbing Reactive Power that are connected at 100 kV or higher, or through a dedicated transformer with a high-side voltage of 100 kV or higher, or through a transformer that is designated in Inclusion I1 unless excluded by application of Exclusion E4 Exclusions: • E1 - Radial systems: A group of contiguous transmission Elements that emanates from a single point of connection of 100 kV or higher and: a) Only serves Load. Or, b) Only includes generation resources, not identified in Inclusions I2, I3, or I4, with an aggregate capacity less than or equal to 75 MVA (gross nameplate rating). Or, c) Where the radial system serves Load and includes generation resources, not identified in Inclusions I2, I3 or I4, with an aggregate capacity of non-retail generation less than or equal to 75 MVA (gross nameplate rating). Note 1 – A normally open switching device between radial systems, as depicted on prints or one-line diagrams for example, does not affect this exclusion. Note 2 – The presence of a contiguous loop, operated at a voltage level of 50 kV or less, between configurations being considered as radial systems, does not affect this exclusion. • E2 - A generating unit or multiple generating units on the customer's side of the retail meter that serve all or part of the retail Load with electric energy if: (i) the net capacity provided to the BES does not exceed 75 MVA, and (ii) standby, back-up, and maintenance power services are provided to the generating unit or multiple generating units or to the retail Load by a Balancing Authority, or provided pursuant to a binding obligation with a Generator Owner or Generator Operator, or under terms approved by the applicable regulatory authority.
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	• E3 - Local networks (LN): A group of contiguous transmission Elements operated at less than 300 kV that distribute power to Load rather than transfer bulk power across the interconnected system. LN's emanate from multiple points of connection at 100 kV or higher to improve the level of service to retail customers and not to accommodate bulk power transfer across the interconnected system. The LN is characterized by all of the following: a) Limits on connected generation: The LN and its underlying Elements do not include generation resources identified in Inclusions I2, I3, or I4 and do not have an aggregate capacity of non-retail generation greater than 75 MVA (gross nameplate rating); b) Real Power flows only into the LN and the LN does not transfer energy originating outside the LN for delivery through the LN; and c) Not part of a Flowgate or transfer path: The LN does not contain any part of a permanent Flowgate in the Eastern Interconnection, a major transfer path within the Western Interconnection, or a comparable monitored Facility in the ERCOT or Quebec Interconnections, and is not a monitored Facility included in an Interconnection Reliability Operating Limit (IROL). • E4 – Reactive Power devices installed for the sole benefit of a retail customer(s). Note - Elements may be included or excluded on a case-by-case basis through b) the Rules of Procedure exception process.
CIP Exceptional Circumstance	A situation that involves or threatens to involve one or more of the following, or similar, conditions that impact safety or BES reliability: a risk of injury or death; a natural disaster; civil unrest; an imminent or existing hardware, software, or equipment failure; a Cyber Security Incident requiring emergency assistance; a response by emergency services; the enactment of a mutual assistance agreement; or an impediment of large scale workforce availability.
CIP Senior Manager	A single official with the authority to bind the registered entity to adhere to the requirements within the CIP Standards, CIP-002 through CIP-015.
Control Center	One or more facilities hosting operating personnel that monitor and control the Bulk Electric System (BES) in real-time to perform the reliability tasks, including their associated data centers, of: 1) a Reliability Coordinator, 2) a Balancing Authority, 3) a Transmission Operator for transmission Facilities at two or more locations, or 4) a Generator Operator for generation Facilities at two or more locations.
Cranking Path	A portion of the electric system that can be isolated and then energized to deliver electric power from a generation source to enable the startup of one or more other generating units.
Cyber Assets	Programmable electronic devices, including the hardware, software, and data in those devices.

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Distribution Provider	Provides and operates the “wires” between the transmission system and the end-use customer. For those end-use customers who are served at transmission voltages, the Transmission Owner also serves as the Distribution Provider. Thus, the Distribution Provider is not defined by a specific voltage, but rather as performing the distribution function at any voltage
Electronic Access Control or Monitoring Systems	Cyber Assets that perform electronic access control or electronic access monitoring of the Electronic Security Perimeter(s) or BES Cyber Systems. This includes Intermediate Systems.
Electronic Access Point	A Cyber Asset interface on an Electronic Security Perimeter that allows routable communication between Cyber Assets outside an Electronic Security Perimeter and Cyber Assets inside an Electronic Security Perimeter.
Electronic Security Perimeter	The logical border surrounding a network to which BES Cyber Systems are connected using a routable protocol.
Element	Any electrical device with terminals that may be connected to other electrical devices such as a generator, transformer, circuit breaker, bus section, or transmission line. An Element may be comprised of one or more components.
External Routable Connectivity	Programmable electronic devices, excluding Shared Cyber Infrastructure, including the hardware, software, and data in those devices. Application containers are considered software of Virtual Cyber Assets (VCAs) or Cyber Assets. VCAs are not considered software or data of Cyber Assets.
Facility	A set of electrical equipment that operates as a single Bulk Electric System Element (e.g., a line, a generator, a shunt compensator, transformer, etc.)
Inoperable or Damaged	The loss of transmission elements preventing the facility from performing its intended function or physical destruction that prevents the facility from being restored to service through normal operating procedures
Interactive Remote Access	User-initiated access by a person employing a remote access client or other remote access technology using a routable protocol. Remote access originates from a Cyber Asset that is not an Intermediate System and not located within any of the Responsible Entity’s Electronic Security Perimeter(s) or at a defined Electronic Access Point (EAP). Remote access may be initiated from: 1) Cyber Assets used or owned by the Responsible Entity, 2) Cyber Assets used or owned by employees, and 3) Cyber Assets
Physical Security Perimeter	The physical border surrounding locations in which BES Cyber Assets, BES Cyber Systems, or Electronic Access Control or Monitoring Systems reside, and for which access is controlled.

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Protected Cyber Assets	One or more Cyber Assets connected using a routable protocol within or on an Electronic Security Perimeter that is not part of the highest impact BES Cyber System within the same Electronic Security Perimeter. The impact rating of Protected Cyber Assets is equal to the highest rated BES Cyber System in the same ESP.
Shared Cyber Infrastructure	One or more programmable electronic devices, including the software that shares the devices' resources, that: • Hosts one or more Virtual Cyber Assets (VCA) included in a BES Cyber Systems (BCS) or their associated Electronic Access Control or Monitoring Systems (EACMS) or Physical Access Control Systems (PACS); and hosts one or more VCAs that are not included in, or associated with, BCS of the same impact categorization; or • Provides storage resources required for system functionality of one or more Cyber Assets or VCAs included in a BCS or their associated EACMS or PACS; and also for one or more Cyber Assets or VCAs that are not included in, or associated with, BCS of the same impact categorization. SCI does not include the supported VCAs or Cyber Assets with which it shares its resources.
Transient Cyber Asset	A Cyber Asset that is: 1. capable of transmitting or transferring executable code, 2. not included in a BES Cyber System, 3. not a Protected Cyber Asset (PCA) associated with high or medium impact BES Cyber Systems, and 4. directly connected (e.g., using Ethernet, serial, Universal Serial Bus, or wireless including near field or Bluetooth communication) for 30 consecutive calendar days or less to a: • BES Cyber Asset, • network within an Electronic Security Perimeter (ESP) containing high or medium impact BES Cyber Systems, or • PCA associated with high or medium impact BES Cyber Systems. Examples of Transient Cyber Assets include, but are not limited to, Cyber Assets used for data transfer, vulnerability assessment, maintenance, or troubleshooting purposes.
Transmission	An interconnected group of lines and associated equipment for the movement or transfer of electric energy between points of supply and points at which it is transformed for delivery to customers or is delivered to other electric systems.
Transmission Facility	A Transmission Facility is a set of electrical equipment (including, but not limited to, lines, transformers, breakers, sectionalizing devices, and associated controls) that: • Operates as a single Bulk Electric System (BES) element for transmission at or above 100 kV; • Is functionally integrated such that an outage or failure at any point within that set of equipment effectively interrupts the same

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	transmission path or paths across the system.
Transmission Line	A system of structures, wires, insulators and associated hardware that carry electric energy from one point to another in an electric power system. Lines are operated at relatively high voltages varying from 69 kV up to 765 kV, and are capable of transmitting large quantities of electricity over long distances.
Transmission Planner	The entity that develops a long-term (generally one year and beyond) plan for the reliability (adequacy) of the interconnected bulk electric transmission systems within its portion of the Planning Authority area.
Transmission Operator	The entity responsible for the reliability of its “local” transmission system, and that operates or directs the operations of the transmission Facilities.
Transmission Owner	The entity that owns and maintains transmission Facilities.